

2c. Content of Pure Core (Mandatory papers Y540 and Y541)

Introduction to Pure Core.

In **Pure Core** learners will extend and deepen their knowledge of proof, algebra, functions, calculus, vectors and differential equations studied in A Level Mathematics. They will also broaden their knowledge into other areas of pure mathematics that underpin the further study of mathematics and other numerate subjects with complex numbers, matrices, polar coordinates and hyperbolic functions.

4.01 Proof

Proof by induction is introduced, including its application in proofs on sums of simple series, powers of matrices and divisibility.

4.02 Complex Numbers

Complex numbers and their basic arithmetic are introduced, including in modulus-argument form. They are used to solve polynomial equations with real coefficients and to define loci on the Argand diagram. De Moivre's theorem is used to develop trigonometrical relationships. Roots of complex numbers are used to help to solve geometrical problems using an Argand diagram.

4.03 Matrices

Matrix arithmetic is introduced and applied to linear transformations in 2-D, and some in 3-D, including the concept of invariance. Determinants and inverses of 2×2 and 3×3 matrices are found and used to solve matrix equations.

4.04 Further Vectors

Vector equations of lines and planes are studied; methods for finding angles and distances between points, lines and planes are developed. Scalar and vector products are introduced, and used in a variety of geometrical problems. How planes intersect in 3-D space is considered, and matrices are used to find the point(s) of intersection.

4.05 Further Algebra

Relationships between roots of and coefficients of polynomials are explored. Techniques involving partial fractions are developed.

4.06 Series

Simple series are summed, using standard formulae and the method of differences. Partial fractions are used to sum series.

4.07 Hyperbolic functions

Hyperbolic functions are introduced and their inverses are used in integration.

4.08 Further Calculus

Maclaurin series are used to approximate functions. Integration techniques are extended to include improper integrals, volumes of revolution, mean values of functions and partial fractions. Inverse trigonometric functions are defined and used for integration.

4.09 Polar Coordinates Curves defined in polar coordinates are explored, including finding the area enclosed by a curve.

4.10 Differential Equations

The work in A Level Mathematics is extended to include the integrating factor method for first order differential equations. The general 2nd order linear differential equation is solved, including SHM and damped oscillations. Simple systems of linear 1st order differential equations are explored.

Assumed knowledge

Learners are assumed to know the content of GCSE (9–1) Mathematics and A Level Mathematics. All of this content is assumed, but will only be explicitly assessed where it appears in this specification.

Use of technology

To support the teaching and learning of mathematics using technology, we suggest that the following activities are carried out through the course:

1. Graphing tools: Learners could use graphing software to investigate the relationships between graphical and algebraic representations, including complex numbers, vectors, implicit and parametric curves, and differential equations.

2. Computer Algebra System (CAS): Learners could use CAS software to investigate algebraic relationships, including derivatives and integrals, to solve equations, including differential equations, to manipulate matrices and as an investigative problem solving tool. This is best done in conjunction with other software such as graphing tools and spreadsheets.
3. Visualisation: Learners could use appropriate software to visualise situations in 3-D relating to lines and planes, and to linear transformations.
4. Spreadsheets: Learners could use spreadsheet software to investigate numerical methods, sequences and series, for modelling and to generate tables of values for functions.

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When this course is being co-taught with OCR's AS Level Further Mathematics A (H235) the 'Stage 1' column indicates the common content between the two specification and the 'Stage 2' column indicates content which is particular to this specification. In each section the OCR Reference lists 'Stage 1' statements before 'Stage 2' statement

OCR Ref.	Subject Content	Stage 1 learners should...	Stage 2 learners should additionally...	DfE Ref.
4.01 Proof				
4.01a 4.01b	Mathematical induction	a) Be able to construct proofs using mathematical induction. <i>This topic may be tested using any relevant content including divisibility, powers of matrices and results on powers, exponentials and factorials.</i> <i>e.g. Prove that $\begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}^n = \begin{pmatrix} 1 & 0 \\ n & 1 \end{pmatrix}$ for $n \in \mathbb{Z}^+$.</i> <i>Prove that $7^n - 3^n$ is divisible by 4 for $n \in \mathbb{Z}^+$.</i> <i>Prove that $2^n > 2n$ for $n \geq 3, n \in \mathbb{Z}$.</i>	b) Be able to construct proofs of a more demanding nature, including conjecture followed by proof. <i>This topic may be tested using any relevant content including sums of series.</i> <i>e.g. Prove that $\sum_{r=1}^n \frac{1}{r(r+1)} = \frac{n}{n+1}$ for all $n \in \mathbb{Z}$.</i> <i>Prove that $3^n > n^3$ for $n \geq 4, n \in \mathbb{Z}$.</i> <i>Prove that $n! < n^n$ for $n \geq 1, n \in \mathbb{Z}$.</i> <i>Prove that $(1+x)^n \geq 1+nx$ for any real number $x > -1$ and $n \in \mathbb{Z}^+$.</i> <i>Prove that the nth derivative of $x^2 e^x$ is $(x^2 + 2nx + n(n-1))e^x$.</i>	A1